

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

BIOS / POST Error Message	Hardware or Configuration Issue
1. CMOS Checksum Error	CMOS battery is weak or BIOS settings have been reset.
2. Keyboard Error or No Keyboard Present	Keyboard is not connected properly or is faulty.
3. No Boot Device Found	Hard Disk/SSD is not detected or the boot device is not configured correctly.
4. Memory Test Failed	RAM is faulty, loose, or not installed correctly.
5. CPU Fan Error	CPU fan is not connected, not working, or spinning too slowly, which may cause overheating.

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

Meaning of the POST Beep Code:

In an **AMI BIOS**, **3 short beeps (repeating)** usually indicate a **Base 64 KB RAM (Memory) Failure**. This means the BIOS has detected a problem with the system memory (RAM), such as a faulty RAM module or a RAM module that is not seated correctly.

Troubleshooting Steps

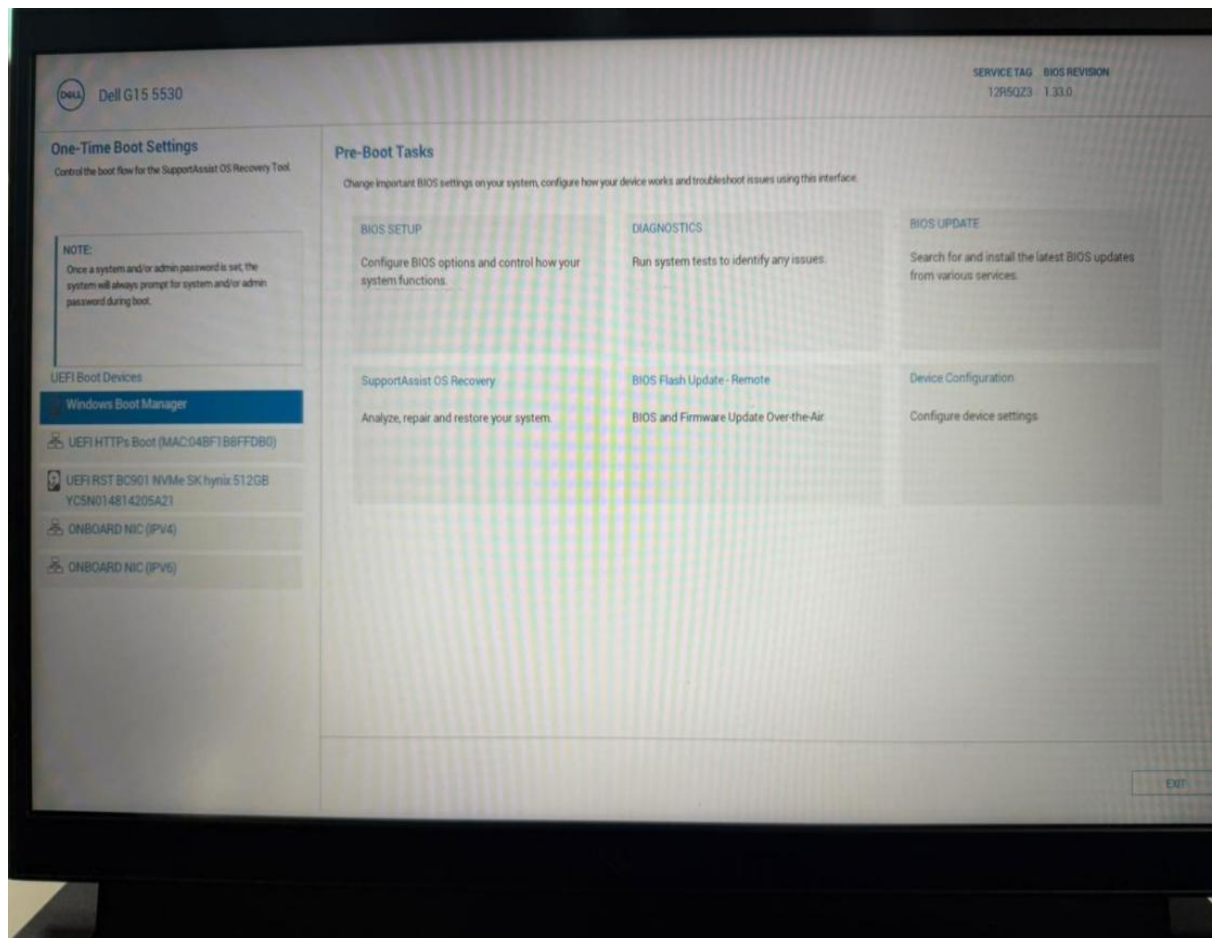
1. Reseat the RAM

- Turn off the PC and unplug the power.
- Remove the RAM module(s), clean the contacts if necessary, and reinstall them firmly in the memory slot.

2. Test the RAM

- Try booting the PC with **one RAM stick at a time** in different memory slots.
- If the system boots with one module but not another, replace the faulty RAM module.

3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset.

CMOS Checksum Error – Defaults Loaded

Meaning:

The message "**CMOS checksum error – Defaults loaded**" means that the BIOS detected that the **CMOS memory data is incorrect or has been reset**. This usually happens because the **CMOS battery is weak or dead**, or the BIOS settings have been cleared. As a result, the BIOS loads its **default settings**.

How to Resolve the Issue

1. **Replace the CMOS Battery**

- Replace the old **CR2032 CMOS battery** with a new one if it is weak or dead.

2. **Enter the BIOS Setup**

- Restart the PC and press the BIOS key (such as **F2**, **Delete**, or **F10**) to enter the BIOS setup.

3. **Set the Correct Date and Time**

- Update the system date and time and reconfigure any required BIOS settings.

4. **Save and Exit**

- Save the BIOS settings and restart the computer.

5. **Check if the Error is Resolved**

- If the error no longer appears, the issue has been fixed. If it continues, the motherboard or CMOS memory may need further inspection.